# STANDATA interpretation 23-BCI-024/23-FCI-008

Building/Fire

## Variance process for industrial occupancy

Date Issued: August 2026



## Purpose

To clarify the variance application, review, and enforcement process for authorities having jurisdiction (AHJs) and designers that may allow for a high-hazard industrial (F-1) occupancy to be classified as a medium-hazard industrial (F-2) occupancy, or for F-1 to be addressed as presenting a similar risk to an F-2.

This also addresses the relevant *Safety Codes Act* (SCA) and Code requirements and is not a design or evaluation guide for the required engineering analysis.

## Discussion

The National Building Code – 2023 Alberta Edition (NBC(AE)) and the National Fire Code – 2023 Alberta Edition (NFC(AE)) establish major occupancy classifications for various types of occupancies or building uses.

In Alberta, designers have challenged the F-1 occupancy classification for certain industrial occupancies. An F-1 occupancy currently does not allow an attached restaurant or beverage establishment within the same building. A variance provides a path by which a restaurant or beverage establishment may be included in the same building as the following occupancies, but not limited to a flour mill, distillery, or cereal mill.

Designers have advocated that the current Codes are outdated or overly restrictive for certain industrial occupancy classifications, compared to building and fire codes in jurisdictions outside of Canada reflecting current industry practices. Designers have asserted that it can be demonstrated their proposed design poses a hazard level equivalent to an F-2 occupancy. AHJs in Alberta have required designers to rely upon an assortment of interpretations and variances to allow for this change in classification.

This patchwork of interpretations and variances has resulted in proposed re-classifications or variance applications being approached in conflicting ways among AHJs. In some cases, AHJs have rejected variance applications from the outset, with the opinion that a variance application cannot seek to change an F-1 to an F-2 occupancy classification, or that the risk analysis cannot demonstrate a similar risk to an F-2 occupancy (See **AHJ Issuance or Refusal of a Variance Application**).

This STANDATA reaffirms the variance application, review, and enforcement process to support consistency across Alberta.

### Engineering Analysis Variance Application

Where an occupancy classification appears to fall between two different major occupancy classifications, the safety codes officer (SCO) typically may use their judgment and discretion to determine the most appropriate classification. In most cases, the SCO can rely on their own experience and judgement to make this determination without the need for a professional engineering analysis or variance process.

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However, in the case of the distinction between the F-1 and F-2 occupancy definitions and their associated risks, it is based on the nature of the hazard, with an F-1 occupancy considered to pose a “special hazard” compared to an F-2 occupancy. This special hazard is due to sufficient quantities of highly combustible, flammable, or explosive materials and their inherent characteristics. Due to the safety concerns and level of complexity involved in assessing any possible special hazard and determining the appropriate occupancy classification, it is necessary for the SCO to require professional involvement (see Sentence 2.4.1.1.(4) of Division C of the NBC(AE)).

While the SCA allows for variances, which may apply to any item related to the administration of the SCA, including the occupancy classification of a building. With the caveat that it is inappropriate to apply a variance to classify an occupancy that is under a fundamentally different use than the Code’s default classification (i.e., classifying a care occupancy as an assembly occupancy), a variance may be appropriate where the use is broadly the same (i.e., industrial) but the nature and level of the hazard are in question.

For this type of variance application, the applicant must assess the fire and explosion hazard risk characteristics of the proposed occupancy and demonstrate through an engineering analysis that no special hazard exists (or the hazard has been appropriately addressed through hazard control systems), such that an approximately equivalent or greater safety performance level relative to an F-2 occupancy has been achieved.

As such, the engineering analysis for such a variance must include the documentation required under Section 2.3., “Alternative Solutions,” of Division C of the NBC(AE).

Section 2.3. of Division C requires that information about the qualifications, experience, and background of the person taking responsibility for the design be included as part of the variance application. If the SCO has any related concerns, they may consult the Association of Professional Engineers and Geoscientists of Alberta (APEGA). APEGA is responsible for regulating individuals and companies that practice engineering or geoscience to ensure they are competent, ethical, and professional in the work they perform, and has the authority to investigate any unprofessional conduct or unskilled practice.

**AHJ Issuance or Refusal of a Variance Application**

If an SCO is satisfied that the conditions required in this STANDATA have been met (see Interpretation section), the SCO should consider it acceptable to issue a variance.

An SCO may refuse to issue a variance if any of the following conditions apply:

- There is insufficient or incomplete information;
- The input parameters or assumptions are incorrect;
- The intent of a variance application is not demonstrated in the analysis (approximately equivalent or greater safety performance); or
- The recommendation is unclear, or impractical/unachievable (i.e. constraints are impossible to achieve or affect other tenancies or buildings).

If an SCO refuses to issue the variance, the applicant must be served with a written notice of the refusal. The applicant may appeal the refusal to the Safety Codes Council within 30 days after the date on which the notice was received (see sections 38(6) and 38(7) of the SCA).

**Ongoing Code Compliance – NFC(AE) Enforcement**

After the design, construction and occupancy have been approved, the fire SCO may conduct ongoing NFC(AE) compliance inspections including provisions related to the following:

- The maintenance and use of the fire safety and fire protection features of the occupancy;
- The conduct of activities that may cause fire hazards in and around the occupancy;
- Limitations on hazardous contents in and around the occupancy.



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In addition to the provisions of the NFC(AE), the fire SCO should be able to confirm that the engineering analysis conditions under which the variance was issued are still being met, including any special maintenance or operational requirements. (See sections 34 and 35 of the SCA and Sentence 2.3.1.2.(1) of Division C of the NFC(AE) concerning inspection and documentation review).

If any changes in use or alterations within the occupancy are intended, including any changes to operations, processes, or equipment, the owner must consult the AHJ to determine if a building permit is required. A variance for an industrial occupancy to allow an F-2 occupancy classification or an F-1 occupancy classification with an attached restaurant or beverage establishment may be invalidated by such changes.

## Safety Codes Act References

***Safety Codes Act* Part 2 Administration**

**Inspections**

34(1) For the purpose of ensuring that this Act and any thing issued under this Act are complied with, a safety codes officer may, without a warrant, at any reasonable time, enter any premises or place, except a private dwelling place that is in use as a dwelling, in which the officer has reason to believe there is something to which this Act applies and may, using reasonable care, carry out an inspection, review designs and examine and evaluate quality management systems and manufacturing and construction processes.

***Safety Codes Act* Part 2 Administration**

**Production of documents**

35(1) For the purpose of ensuring that this Act and any thing issued under this Act are complied with, a safety codes officer may demand the production, within a reasonable time, of any record or document pertaining in any manner to compliance with this Act and may on giving a receipt for it remove it for not more than 48 hours for the purpose of making copies of it.

***Safety Codes Act* Part 3 Standards**

**Variances**

38(1) An Administrator or a safety codes officer may issue a written variance with respect to any thing, process or activity to which this Act applies if the Administrator or officer is of the opinion that the variance provides approximately equivalent or greater safety performance with respect to persons and property as that provided for by this Act.

(2) An Administrator or a safety codes officer may include terms and conditions in the variance.

(3) A safety codes officer on issuing a variance shall notify an Administrator.

(4) The Regulations Act does not apply to variances issued under this section.

(5) A decision of the Administrator under subsection (1) is final and not subject to appeal.

(6) If a safety codes officer refuses to issue a written variance, the applicant shall be served with a written notice of the refusal.



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(7) An applicant who receives a notice referred to in subsection (6) may, within 30 days after the date on which the notice is received, appeal the refusal to the Council in accordance with the Council's bylaws.

***Safety Codes Act* Part 5 Orders, Appeals**

**Appeal of refusals, suspensions, cancellations**

51(1) The Council, on receipt of a notice of appeal with respect to

(a) a refusal to designate a corporation as an accredited corporation or a person as an accredited agency,

(a.1) a refusal by a safety codes officer to issue a written variance,

(b) a refusal to register a design or a deregistration of a design, or

(c) a suspension or cancellation of a designation of accreditation, a certificate of competency or a permit,

shall send a copy of the notice of appeal to the relevant Administrator and the safety codes officer, if any, who issued the suspension or cancellation, and notify them and the appellant of the time and place of the appeal.

(2) In order for an appeal to proceed, the Council must receive a notice of appeal in the form approved by the Council within 30 days after the date the corporation or person was served with the written notice of the refusal to designate, refusal by a safety codes officer to issue a written variance, refusal to register, deregistration, suspension or cancellation.

**Code References**

**NBC(AE) Division A, Article 1.4.1.2.**

**1.4.1.2. Defined Terms**

*Authority having jurisdiction* means a safety codes officer in the building discipline exercising authority pursuant to designation of powers and terms of employment in accordance with the Safety Codes Act.

*Building* means any structure used or intended for supporting or sheltering any use or *occupancy*.

*Combustible* means that a material fails to meet the acceptance criteria of CAN/ULC-S114, "Standard Method of Test for Determination of Non-Combustibility in Building Materials."

*Floor area* means the space on any *storey* of a *building* between exterior walls and required *firewalls*, including the space occupied by interior walls and *partitions*, but not including *exits*, *vertical service spaces*, and their enclosing assemblies.

*High-hazard industrial occupancy (Group F, Division 1)* means an industrial occupancy containing sufficient quantities of highly combustible and flammable or explosive materials which, because of their inherent characteristics, constitute a special fire hazard.

*Licensed interior designer* means an individual who is authorized to engage in that portion of the practice of architecture that is defined as interior design under the Architects Act and its Regulations.

*Medium-hazard industrial occupancy* (Group F, Division 2) means an *industrial occupancy* in which the *combustible* content is more than 50 kg/m$^2$ or 1 200 MJ/m$^2$ of *floor area* and not classified as a *high-hazard industrial occupancy*.



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*Registered architectural professional* means an individual who is authorized to engage in the practice of architecture under the Architects Act and its Regulations.

*Registered engineering professional* means an individual who is authorized to engage in the practice of engineering under the Engineering and Geoscience Professions Act and its Regulations.

*Registered professional* means an individual who qualifies as a
(a) *registered architectural professional*,
(b) *registered engineering professional*, or
(c) *licensed interior designer*.

**NBC(AE) Division C, Section 2.3.**

**Section 2.3. Alternative Solutions**
**2.3.1. Documentation of Alternative Solutions**
(See Note A-2.3.1.)

**2.3.1.1. Documentation**
1) Documentation conforming to this Subsection shall be provided by the person requesting the use of an alternative solution to demonstrate that the proposed alternative solution complies with this Code.

2) The documentation referred to in Sentence (1) shall include
a) a Code analysis outlining the analytical methods and rationales used to determine that a proposed alternative solution will achieve at least the level of performance required by Clause 1.2.1.1.(1)(b) of Division A, and
b) information concerning any special maintenance or operational requirements, including any *building* component commissioning requirements, that are necessary for the alternative solution to achieve compliance with the Code after the *building* is constructed.

3) The Code analysis referred to in Clause (2)(a) shall identify the applicable objectives, functional statements and acceptable solutions, and any assumptions, limiting or restricting factors, testing procedures, engineering studies or *building* performance parameters that will support a Code compliance assessment.

4) The Code analysis referred to in Clause (2)(a) shall include information about the qualifications, experience and background of the person or persons taking responsibility for the design.

5) The information provided under Sentence (3) shall be in sufficient detail to convey the design intent and to support the validity, accuracy, relevance and precision of the Code analysis.

6) Where the design of a *building* includes proposed alternative solutions that involve more than one person taking responsibility for different aspects of the design, the applicant for the *permit* shall identify a single person to co-ordinate the preparation of the design, Code analysis and documentation referred to in this Subsection.

**NBC(AE) Division C, Sentence 2.4.1.1.(4)**

**2.4.1. Professional Involvement**
**2.4.1.1. General**
4) If the size or complexity of a *project* may give rise to special safety concerns, the *authority having jurisdiction* may require the involvement of one or more *registered professionals* to authenticate all or part of the drawings and specifications and perform *field reviews*.



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**NFC(AE) Division C, Sentence 2.3.1.2.(1)**

**2.3.1.2. Retention of Documentation**
1) Documentation conforming to Article 2.3.1.1. shall be retained on the premises to which it pertains for each measure that has been approved by the *authority having jurisdiction* and implemented as an alternative solution to acceptable solutions
a) in this Code, or
b) in the NBC(AE) that are attributed to the objectives Fire Safety (OS1), Safety in Use (OS3), Fire Protection of the Building (OP1), or Protection of Adjacent Buildings from Fire (OP3).

## Application

This interpretation applies to a building or part thereof with an F-1 occupancy classification under the NBC(AE) or NFC(AE).

## Interpretation

1. An AHJ may accept the occupancy classification of an F-1 occupancy as an equivalent risk to an F-2 occupancy as part of a variance application, provided that the following conditions are met:

    - An engineering analysis demonstrates that no special hazard exists, or that the special hazard is appropriately addressed through hazard control systems, such that a safety performance level approximately equivalent to or greater than an F-2 occupancy is achieved.
    - The engineering analysis is performed by a registered engineering professional (as defined by the NBC(AE)) who is authorized and qualified to engage in such work.
    - The documentation submitted to support the variance is in conformance with Section 2.3. of Division C of the NBC(AE).
    - Applicable terms and conditions are provided in support of the variance application.

2. To address ongoing Code compliance following occupancy, the owner must have the variance documentation available on the premises for review by the fire SCO, to confirm that the engineering analysis conditions under which the variance was issued have not been contravened.

This INTERPRETATION is applicable throughout the Province of Alberta.

Unless stated otherwise, all Code references in this STANDATA are to the National Building Code – 2023 Alberta Edition or National Fire Code – 2023 Alberta Edition

Issued by the Provincial Building and Fire Administrators

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